

EDHUBILLI SANDEEP KUMAR

B.Tech Information Technology • NLP & AI Systems • Full-Stack Engineering • Human-Computer Interaction

+91 7569823540 • edhubillisandeepkumar1234@gmail.com • github.com/sandeep-kumar-270904 • linkedin.com/in/edhubilli-sandeep-kumar-240407392 • sandeep-kumar-edhubilli-portfolio.lovable.app • Visakhapatnam, India

RESEARCH INTEREST & SUMMARY

B.Tech Information Technology student (ANITS, 2026) with hands-on experience in Natural Language Understanding, real-time collaborative systems, and full-stack software engineering. Independently studied NLP, LLM, and conversational AI research literature; applied findings to build production-grade systems. Proficient in Python, JavaScript, and Java. Delivered three original projects spanning NLP pipelines, distributed web platforms, and real-time multi-user collaboration systems — demonstrating research-oriented engineering and a drive to solve real problems at scale.

RESEARCH INTERESTS

Natural Language Understanding • Conversational AI & Dialogue Systems • Human-Computer Interaction • Real-Time Collaborative Systems • Machine Learning • Software Engineering at Scale

TECHNICAL SKILLS

Languages: Python • JavaScript • TypeScript • Java • C/C++ (basic) • SQL • PLpgSQL • HTML5 • CSS3

AI / ML & NLP: Intent Classification • NLP Pipelines • scikit-learn • LangChain • Hugging Face • TensorFlow (basics)

Web & Backend: React.js • Node.js • Express.js • Flask • Django • REST APIs • WebSockets • Supabase • TanStack Query

Databases & Infra: PostgreSQL • MongoDB • MySQL • Redis • Docker • Vercel • GitHub Actions • Cypress E2E

Tools: Git • GitHub • VS Code • Figma • Jupyter Notebook • Vite • shadcn/ui • Framer Motion

PROJECTS & RESEARCH WORK

CollabSphere — Real-Time Student Collaboration Platform | [TypeScript](#) • [React](#) • [Supabase](#) • [PostgreSQL](#) • [WebSockets](#) • [Cypress](#) • [Vercel](#)

- Architected a full-stack idea management and collaboration platform enabling students to post ideas, request to join teams, and receive real-time acceptance notifications — zero manual page refreshes via WebSocket-powered Supabase Realtime.
- Engineered WebSocket subscription layer with automatic fallback to polling, live sync indicators, and TanStack Query cache management — maintaining sub-second UI propagation across all connected clients simultaneously.
- Implemented Row-Level Security (RLS) policies in PostgreSQL, JWT-based multi-user authentication, and PLpgSQL custom database functions; deployed on Vercel with 220+ commits demonstrating sustained engineering discipline.
- Wrote Cypress E2E test suite covering the full Innovation Hub ↔ Dashboard real-time loop — validating zero-refresh synchronization, collaboration counter accuracy, and sync status correctness under concurrent user load.

ANITS Academic NLP Assistant — Multilingual Dialogue System | [Python](#) • [NLP](#) • [Intent Classification](#) • [Flask](#) • [React.js](#) • [MySQL](#)

- Designed and implemented an NLP-based conversational system enabling students to query academic schedules, events, and FAQs in natural language — independently studied intent classification research to inform the pipeline architecture.
- Built custom tokenization, feature extraction, and multi-class intent classification pipeline covering 10+ dialogue categories; documented evaluation methodology and improvement hypotheses in research-style project writeup.
- Deployed a Flask REST API with MySQL session persistence and React.js real-time chat frontend; applied findings from NLP literature to reduce misclassification on ambiguous queries.

CineScore — Distributed Movie Review & Recommendation API | [Node.js](#) • [PostgreSQL](#) • [Redis](#) • [Docker](#) • [JWT](#) • [GitHub Actions](#)

- Built a production-ready REST API platform with weighted rating algorithms, collaborative filtering recommendations, multi-stage moderation workflow, and fraud/bot detection using behavioral pattern analysis.
- Achieved sub-100ms API response times via Redis caching (TTL-managed); containerized with Docker, deployed CI/CD pipeline via GitHub Actions, and configured Nginx reverse proxy — demonstrating DevOps and software engineering at scale.

EXPERIENCE

Web Developer Intern — Inoninks

2024

Developed and shipped responsive full-stack web features; contributed to REST API integration, cross-browser compatibility testing, and collaborative engineering workflows in a professional team environment.

EDUCATION

B.Tech in Information Technology — ANITS, Visakhapatnam

2022 – Present (3rd Year)

CGPA: 7.4 / 10.0 • Affiliated to Andhra University • NAAC 'A+' Grade

Relevant Coursework: Data Structures & Algorithms • DBMS • Object-Oriented Analysis & Design • Software Engineering • Web Technologies • Probability & Statistics

RESEARCH ENGAGEMENT & LEADERSHIP

- Independent Research Reading: Regularly studied published papers and technical blogs on NLP, large language models, conversational AI, and real-time systems — directly applied findings to architecture decisions in personal projects.

- Hackathon Participant: Delivered functional AI/web prototypes under research-driven time-constrained problem statements across multiple college and inter-college hackathon events.
- NSS Secretary — ANITS: Directed a 100+ member chapter; orchestrated cross-team logistics for 10+ social initiatives — demonstrating leadership, communication, and organizational scale.
- Campus Club Social Media & Content Handler: Produced technical content and managed digital community engagement — communicating complex engineering ideas to diverse audiences.

CERTIFICATIONS

- Machine Learning Specialization — Andrew Ng · Coursera (Stanford University) · In Progress
- Docker & Container Orchestration — KodeKloud / Udemy
- GitHub Actions CI/CD — GitHub Learning Lab